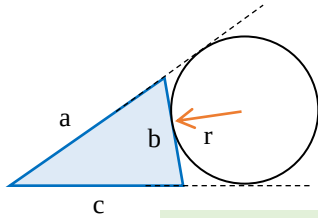




Continuation... Area of Triangles

6. Area of with escribed circle



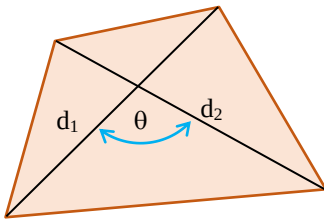
$$A = r(s - b)$$

$$s = \frac{a + b + c}{2}$$

Note: b is the side physically tangent to the circle

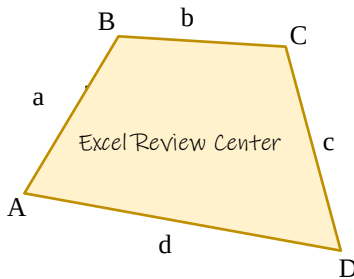
Area of Quadrilaterals

2. Given 2 diagonals and an included angle



$$A = \frac{1}{2} d_1 d_2 \sin \theta$$

1. Given 4 sides and 2 opposite angles



$$A = \sqrt{(s-a)(s-b)(s-c)(s-d) - abcd \cos^2 \theta}$$

$$s = \frac{a + b + c + d}{2}$$

$$\theta = \frac{A + C}{2}$$

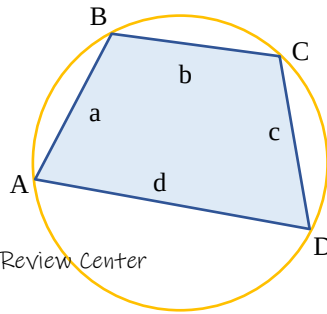
$$\theta = \frac{B + D}{2}$$



Did You Know?

Excel Review Center was the first review center to introduce computer-aided review classes in 1995!

3. Cyclic quadrilateral (All vertices lie on a circle)



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$$A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

$$s = \frac{a + b + c + d}{2}$$

$$A + C = 180^\circ$$

$$B + D = 180^\circ$$

Ptolemy's Theorem:

"The sum of the two pairs of opposite sides of a convex quadrilateral inscribed in a circle is equal to the product of the lengths of the diagonals".

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$$ac + bd = d_1 d_2$$

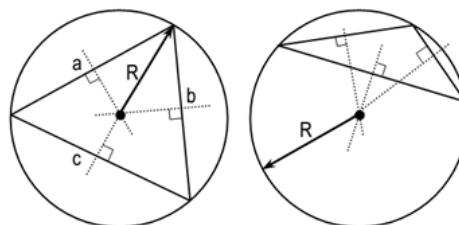
This theorem was named after the geographer, mathematician and astronomer, Ptolemy or Claudius Ptolemaeus (c.100 – c.168 A.D.) of Alexandria.

Important Points in a Triangle

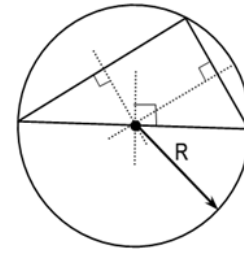
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Circumcenter – the point of concurrency of the perpendicular bisectors of the sides of a triangle.

The circumcenter is also the center of the circumscribing circle (also known as circumcircle). The circumcenter can be inside or outside the triangle.



For right triangle, the circumcenter is at the midpoint of the hypotenuse.

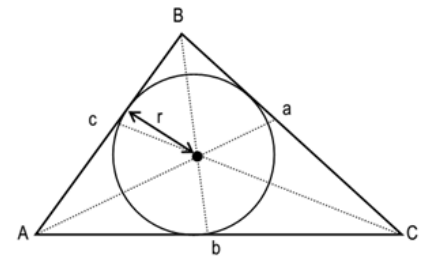


The radius of circumscribing circle is obtained using the following formula:

$$R = \frac{abc}{4A_{\text{triangle}}}$$

Incenter – the point of concurrency of the angle bisector of a triangle. The point of intersection of the angle bisectors of a triangle is called the incenter, which is the center of the inscribed circle (also known as incenter).

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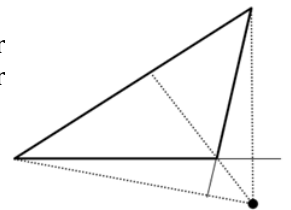


The distance from the incenter to any side of a triangle is equal to the radius of the inscribed circle and can be calculated using the following formula:

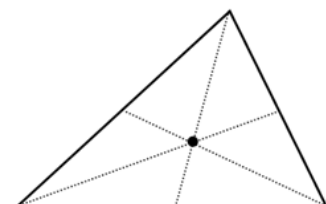
$$r = \frac{A_{\text{triangle}}}{s}$$

$$s = \frac{a + b + c}{2}$$

Orthocenter – the point of concurrency of the altitudes of a triangle. Like circumcenter, the orthocenter can be inside or outside of the triangle.



Centroid – the point of concurrency of the medians of the triangle. Centroid is the geometric center of the triangle. The term "median" refers to the line that connects the vertex to the midpoint of the line opposite the vertex. The centroid is also called the "median point".



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